CS224v

Conversational Virtual Assistants with Deep Learning

Lecture 9: Task-Oriented Agents

Monica Lam and Harshit Joshi

BBC

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Airline held liable for its chatbot giving passenger bad advice - what this means for travellers

23 February 2024

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Maria Yagoda
Features correspondent

Air Canada's chatbot promised a discount
that wasn't available to passenger Jake Moffatt.

The British Columbia Civil Resolution Tribunal ruled that
Air Canada had to pay Moffatt $812.02 (£642.64) in damages and tribunal fees.

A Case Study
Course Enrollment at Stanford

Let's consider a CS MS student

Enrollment Task: invoke enroll

```
from typing import TypedDict
from enum import Enum

class GradeType(Enum):
    LETTER = "Letter"
    CRED_NOCRED = "Cred_noCred"

class CourseToTake(TypedDict):
    course_name: str
    grade_type: GradeType
    num_units: int
    confirm: bool
```

```
class StudentInfo(TypedDict):
    student_name: str
    student_ID: str
    student_email_address: str

def enroll (student_info: StudentInfo,
            course_to_take: CourseToTake)
    -> None:
    pass
```

A simple enough task, or is it?

Course Enrollment Agent

What course would you like to enroll in?

User:

I am studying computer science.
I want to complete the significant implementation requirement.
What are some of the options that require the least amount of work?

A Course Enrollment Assistant

Course Enrollment Agent

What course would you like to enroll in?

Students need to consult:

Degree Requirements

- Unique to each department and specialization
- Program sheets

Stanford | Bulletin ExploreCourses

- All offered courses
- Description, instructors, offerings, and units

CARTA

- Ratings, reviews, hours of work, enrollment outcomes, and sequencing

AI

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Artificial Intelligence Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email: _____ Proposed date for degree conferral: _____ HCP? _____ Coterm? _____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psguide2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.) _____

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279, 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181W, 182/W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B, AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

ARTIFICIAL INTELLIGENCE DEPTH

All depth courses must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS 399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

a) CS 221 (students with equivalent course work may waive with advisor approval)

b) At least four of: CS 223A, 224N, 224R, 224S, 224U, 224V, 224W, 228, 229, 231A, 231N, 234, 237A, 237B, 238

c) A total of at least 21 units from categories (a), (b) and the following: CS 205L, 224C, 224R, 225A, 227B, 229B, 229S, 229T, 230, 233, 235, 236, 239, 246, 257, 270, 271, 173A/273A, 273B, 274, 275, 279, 281, 293, 322, 324, 324H, 325B, 326, 327A, 328, 329 (any suffix), 330, 331, 332, 333, 336, 348I, 348N, 353, 361, 368, 371, 375, 377† (any suffix), 379† (any suffix), 398, 399†, 428A, 428B, 432; EE 263, 276, 278, 364A, 364B, 377; ENGR 205, 209A; MS&E 226, 252; PSYCH 209; STATS 202, 315A, 315B (courses with † notation require approval of MS advisor)

Advisor Notes

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL DEPTH UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.**** _____

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS _____

TOTAL UNITS APPLIED TO MSCS _____

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a B (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

*All classes taken Spring 19-20 and Fall through Summer 20-21 for a CR or S grade will satisfy MSCS requirements as if taken for a letter grade.

Advisor's signature: _____ Date: _____

Computational Biology

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Computational Biology Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email: _____ Proposed date for degree conferral: _____ HCP? _____ Coterm? _____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psgunde2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.) ☐

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279, 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181W, 182W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B; AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

COMPUTATIONAL BIOLOGY DEPTH

All depth courses must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS 399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

a) One of: CS 173A, 273A (students with equivalent coursework may waive with the approval of their advisor)
b) CS221
c) At least two of: CS 142 (or 147L or 193X), 145, 246, 448B
d) At least three of: CS 279, 371; BIOMEDIN 210, 214, 215, 217, 219, 220, 222, 260, 273B; IMMUNOL 207
e) A total of at least 21 units from categories (a), (b), (c), (d) and the following: CS 124, 131, 147, 148, 154, 166, 168, 185, 224N, 224W, 228, 229, 229B, 229S, 229T, 230, 231N, 234, 238, 248/248A, 353, 399; BIO 183, 187; STATS 215, 256

Advisor Notes

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL SPECIALIZATION UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.**** ☐

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS ☐

TOTAL UNITS APPLIED TO MSCS ☐

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a B (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

*All classes taken Spring 19-20 and Fall through Summer 20-21 for a CR or S grade will satisfy MSCS requirements as if taken for a letter grade.

Advisor's signature: _____ Date: _____

Computer and Network Security

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Computer and Network Security Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email: _____ Proposed date for degree conferral: _____ HCP? _____ Coterm? _____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psguide2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.) ☐

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279, 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181/W, 182/W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B; AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

COMPUTER AND NETWORK SECURITY DEPTH

All depth course must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS 399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

a) All of: CS 140/212 (or 140E or 112),† 144†, 155, 255, 356 (students with equivalent course work may waive † flagged courses with advisor approval)

b) At least three of: CS 142, 152, 190, 240 (any suffix), 244, 244B, 249I, 253, 261, 265, 340 (any suffix), 355

c) A total of at least 21 units from categories (a), (b) and the following: CS 245, 251, 294S‡, 399‡; EE 384S (courses with ‡ notation require approval of MS advisor)

Advisor Notes

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL SPECIALIZATION UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.**** ☐

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS ☐

TOTAL UNITS APPLIED TO MSCS ☐

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a B (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

*All classes taken Spring 19-20 and Fall through Summer 20-21 for a CR or S grade will satisfy MSCS requirements as if taken for a letter grade.

Advisor's signature: _____ Date: _____

Software Theory

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Software Theory Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email:_____ Proposed date for degree conferral: _____ HCP?_____ Coterm?_____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psguide2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.)

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279, 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181W, 182/W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B; AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

SOFTWARE THEORY DEPTH

All depth course must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

Advisor Notes

a) CS 242, 243
b) At least one of: CS 221, 244, 245, 341
c) At least one of: CS 255, 350, 355, 356
d) A total of at least 21 units chosen from categories (a), (b), (c) or the following: CS 151, 250, 261, 265, 294S, 295, 315B, 349H, 357, 357S, 399† (courses with † notation require approval of MS advisor)

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL SPECIALIZATION UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.****

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS

TOTAL UNITS APPLIED TO MSCS

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a B (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

*All classes taken Spring 19-20 and Fall through Summer 20-21 for a CR or S grade will satisfy MSCS requirements as if taken for a letter grade.

Advisor's signature: _____ Date:_____

Systems

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Systems Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email: _____ Proposed date for degree conferral: _____ HCP? _____ Coterm? _____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psguide2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.) ____

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279. 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181W, 182/W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B; AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

SYSTEMS DEPTH

All depth course must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

Advisor Notes

a. All of: CS140/212 (or 140E or 112 or 112E) ‡, CS 144‡, 240
b. At least four of: CS 190, 242, 243, 244, 245, 248/248A, 348B; EE 271, 282
c. A total of at least 21 units from categories (a), (b) and the following: CS 149, 217, 229S, 241, 244B, 246, 251, 255, 270, 272, 294S†, 295, 315B, 316, 340 (any suffix), 343D, 344 (any suffix), 348A, 348C, 348E, 348I, 348K, 349 (any suffix and 3 or more units), 356, 357S, 399†, 448 (any suffix); EE 267, 273, 382C, 384A, 384C, 384S (courses with † notation require approval of MS advisor)

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL SPECIALIZATION UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.**** ____

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS ____

TOTAL UNITS APPLIED TO MSCS ____

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a B (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

*All classes taken Spring 19-20 and Fall through Summer 20-21 for a CR or S grade will satisfy MSCS requirements as if taken for a letter grade.

Advisor's signature: _____ Date: _____

Theory

DEPARTMENT OF COMPUTER SCIENCE
MSCS Program Sheet (2023-24)
Theoretical Computer Science Specialization

Name: _____ Advisor: _____ Date: _____

Student ID#: _____ Email: _____ Proposed date for degree conferral: _____ HCP? _____ Coterm? _____

GENERAL INSTRUCTIONS: Before the end of your first quarter, complete a program sheet by filling in the number, name and units of each course you intend to take for your degree. Program sheets, as well as foundation course waiver forms, may be submitted electronically on the Gates Information Network at http://gin.stanford.edu/ using your CS ID and password (click on the 'Dashboard' link). See http://cs.stanford.edu/degrees/mscs/programsheets/psguide2324.pdf for detailed instructions.

FOUNDATIONS REQUIREMENT: You must satisfy the requirements listed in each of the following areas; all courses taken elsewhere must be approved by your advisor via the foundation course waiver form. Do not enter anything in the "Units" column for courses taken elsewhere or for Stanford courses counted towards your undergraduate degree.

Required:	**Equivalent elsewhere** (course numbers/titles/institution)	**Grade**	**Units**
Logic, Automata & Complexity (CS103)			
Probability (CS109, Stat116, CME106, MS&E220, or EE178)			
Algorithmic Analysis (CS161)			
Computer Organ & Sys (CS107 or 107E)			
Principles of Computer Systems (CS111)			

TOTAL UNITS USED TO SATISFY FOUNDATIONS REQUIREMENT (May not exceed **10** units.) _____

SIGNIFICANT IMPLEMENTATION REQUIREMENT: At least one course in your MS program should be designated as satisfying the Significant Implementation Requirement. Note that this course will also be used to satisfy some other requirement (i.e., Depth, or Elective). Must be taken for a letter grade,* must be taken at Stanford. Coterm students who took two of these courses at Stanford as undergraduates may waive this requirement. Deviations must be approved by the faculty director of the MS program, Cynthia Lee.

Course Number:	**Title:**	**Grade**

Courses that may be used to satisfy the significant implementation requirement include: CS 140, 140E, 143, 144, 145, 148, 151, 190, 210B, 212, 224R, 227B, 231N, 239, 243, 248A, 248B, 330, and 341.

BREADTH REQUIREMENT: Three courses, with each course chosen from a different Breadth area A, B, C or D. Breadth courses may not be waived, must be taken for at least 3 units each, and must be completed for a letter grade.* Each of the three Breadth courses must be from different Areas. Note that these courses will also be used to satisfy some other requirement (i.e., Depth or Elective). Letter grade only.*

Course Number:	**Title:**	**Grade**

Area A. Formal Foundations: CS 154, 157, 166, 168, 205L, 229T, 250, 254, 254B, 255, 256, 257, 258, 259Q, 261, 263, 265, 269I, 328, 331, 334A, 354, 355, 359D, 369O; EE 364A, 364B; Phil 251

Area B. Learning and Modeling: CS 173A, 205L, 221, 223A, 224N, 224R, 224S, 224U, 224V, 224W, 227B, 228, 229, 229B, 229T, 230, 231A, 231N, 233, 234, 236, 237A, 237B, 248A, 248B, 271, 272, 273A, 273B, 274, 279, 281, 326, 329D, 329X, 330, 332, 333, 348A, 348B, 348C, 348E, 348I, 348N, 371, 373

Area C. Systems: CS 112 (with CS111 as prereq), 140E, 143, 144, 145, 149, 155, 212, 240, 240LX, 242, 243, 244, 244B, 245, 246, 249I, 316, 348K, 356, 358; EE 180, 282, 382E

Area D. People and Society: CS 147, 148, 152, 181W, 182W, 194H, 206, 247 (any suffix), 256, 269I, 278, 281, 329T, 329X, 347, 377 (any suffix), 384, 448B; AMSTUD 145; ANTHRO 132C, 132D; COMM 254, 266, 286, 324; DESIGN 255; EARTHSYS 213; EDUC 315A; ENGLISH 184D, HISTORY 244F; INTLPOL 268; LAW 4039; ME 177; MS&E 193, 231, 234, 254; POLISCI 150A; PUBLPOL 103F for 3 units, 353B

THEORETICAL COMPUTER SCIENCE DEPTH

All depth course must be taken for a letter grade* for 3 or more units. A maximum of 6 units of CS 399 Independent Study may be counted towards the depth. Any deviations from the stated requirements must be noted and approved by your advisor in the Advisor Notes box provided (maximum of one advisor-approved deviation allowed). Courses taken for your Stanford undergraduate degree do not need to be repeated. Enter course selection in the table below:

Advisor Notes

a) CS 154‡, 261 (students with equivalent course work may waive ‡ courses with advisor approval)

b) A total of at least 21 units from category (a) and the following: CS 151, 163, 166, 168, 228, 229T, 233, 236, 246, 250, 251, 254, 254B, 255, 256, 257, 259Q, 263, 265, 269I, 328, 331, 351, 354, 355, 358, 359 (any suffix)†, 368, 369 (any suffix)†, 399†, 468†; EE 364A; MS&E 310, 319 (courses with † notation require approval of MS advisor)

Note: Multiple CS359, CS369, and/or CS468 courses may be taken as long as they are each on different topics (denoted by different letter suffixes).

Course number	**Title (Depth courses must be taken for at least 3 units.)**	**Grade**	**Units**

TOTAL SPECIALIZATION UNITS APPLIED TO MSCS (must total at least 21 units) ***Letter grades only.**** _____

ELECTIVES

List here any additional courses used to complete the 45-unit requirement for the MSCS degree. You may count up to a maximum of **3 units** of 1-2 unit seminars offered in the School of Engineering. All other electives must be taken for 3 or more units. CS courses numbered above 111, excluding CS 161A, 196, CS198, and CS390A/B/C, can be used as MS electives. Non-CS courses must be technical courses numbered above 100, related to the degree program, and approved by the advisor and the MS program administrator. *Note that CS 129 may not be counted towards the MS if CS 229 is being counted towards any BS or MS requirement.*

Course number	**Title**	**Grade**	**Units**

TOTAL ELECTIVE UNITS APPLIED TO MSCS _____

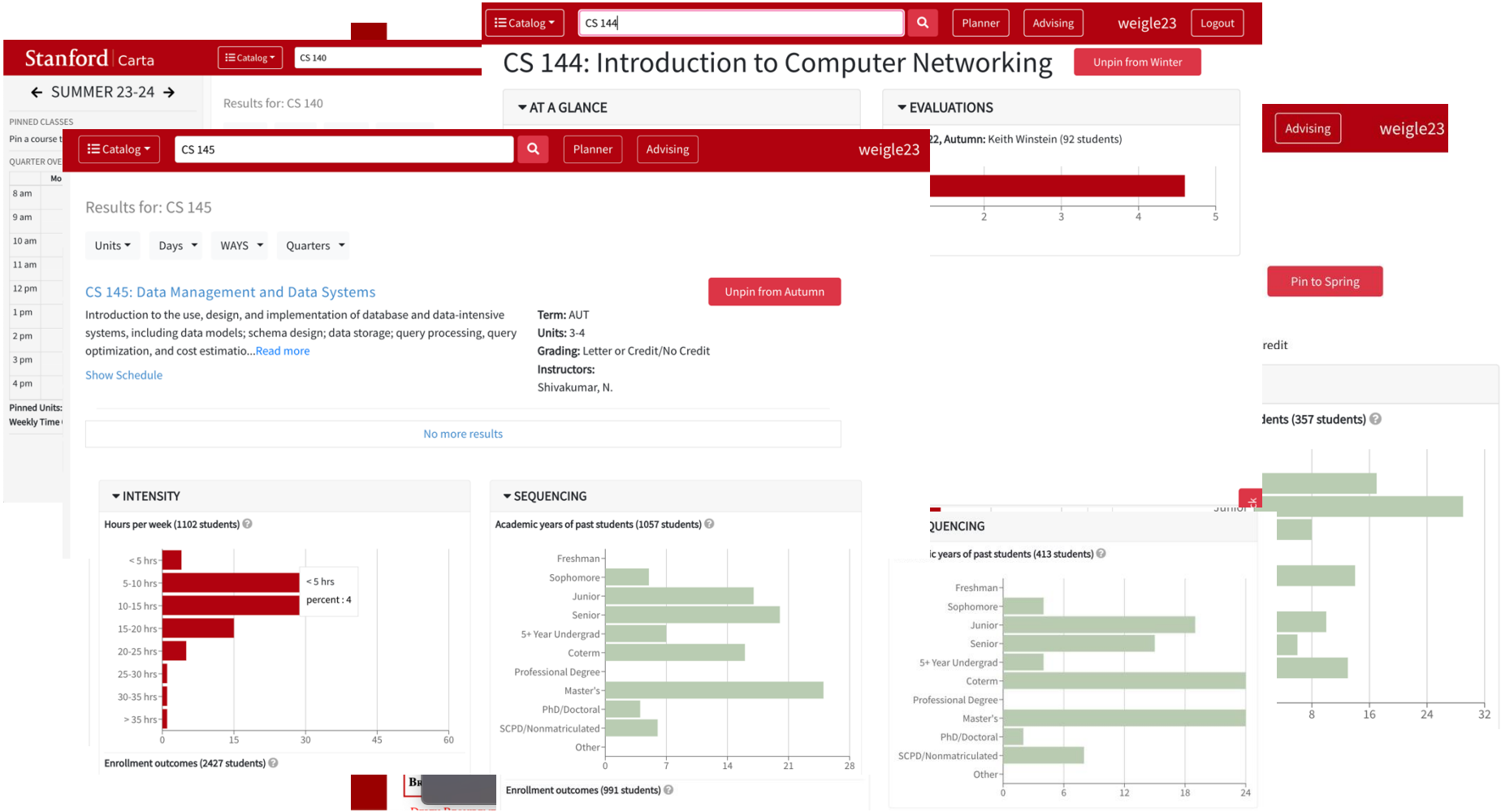
TOTAL UNITS APPLIED TO MSCS _____

ADDITIONAL REQUIREMENTS

- All courses submitted for the MSCS degree must be numbered 100 or greater.
- At most 10 units of Foundations requirement courses may be counted toward your 45 units.
- At most 3 units of 1-2 unit seminars may be counted toward your 45 units.
- At least 36 units submitted for the MSCS degree, including all courses taken for breadth and depth, must be taken for a letter grade.*
- The average grade in the courses submitted for the MSCS must be at least a H (3.0 in Stanford's GPA scale).
- Units previously applied toward BS requirements may not also be counted toward the MSCS.
- You must complete at least 45 graduate units at Stanford before receiving the MSCS degree.

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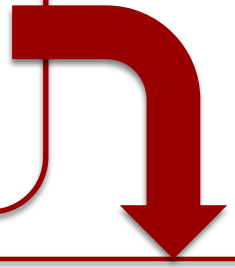
Advisor's signature: _____ Date: _____


Catalog
CS 144
Planner
Advising
weigle23
Logout
Stanford | Carta
Catalog
CS 140
CS 144: Introduction to Computer Networking
Unpin from Winter
SUMMER 23-24
PINNED CLASSES
Pin a course t
QUARTER OVE
Mo
8 am
9 am
10 am
11 am
12 pm
1 pm
2 pm
3 pm
4 pm
Pinned Units:
Weekly Time
Results for: CS 140
AT A GLANCE
EVALUATIONS
Catalog
CS 145
Planner
Advising
weigle23
Advising
weigle23
Results for: CS 145
Units
Days
WAYS
Quarters
CS 145: Data Management and Data Systems
Unpin from Autumn
Introduction to the use, design, and implementation of database and data-intensive systems, including data models; schema design; data storage; query processing, query optimization, and cost estimatio...Read more
Show Schedule
Term: AUT
Units: 3-4
Grading: Letter or Credit/No Credit
Instructors:
Shivakumar, N.
Pin to Spring
redit
No more results
INTENSITY
Hours per week (1102 students)
< 5 hrs
5-10 hrs
10-15 hrs
15-20 hrs
20-25 hrs
25-30 hrs
30-35 hrs
> 35 hrs
< 5 hrs
percent : 4
0
15
30
45
60
Enrollment outcomes (2427 students)
SEQUENCING
Academic years of past students (1057 students)
Freshman
Sophomore
Junior
Senior
5+ Year Undergrad
Coterm
Professional Degree
Master's
PhD/Doctoral
SCPD/Nonmatriculated
Other
0
7
14
21
28
Enrollment outcomes (991 students)
QUENCING
ic years of past students (413 students)
Freshman
Sophomore
Junior
Senior
5+ Year Undergrad
Coterm
Professional Degree
Master's
PhD/Doctoral
SCPD/Nonmatriculated
Other
0
6
12
18
24
dents (357 students)
8
16
24
32

Task assistant are also knowledge assistants

Translates Questions into SUQL

I am a sophomore studying computer science. I want to complete the significant implementation requirement.
What are some of the options that require the least amount of work?



```
SELECT title, course_codes, description,
average_hours_spent
FROM courses WHERE
significant_implementation_requirement = TRUE
ORDER BY average_hours_spent LIMIT 5;
```

Preview of Real User Evaluation

Goal Completion Rate (%)

	Restaurant	Course Enrollment	Service Now Ticket Submission
GPT4: function calling + Genie DB parser	**54.5**	**10.0**	**0.0**
Genie Worksheet	**91.6**	**80.0**	**80.0**

GPT-4 results are not acceptable

- Low completion rate: even 54.5% on restaurant is not good enough!
- Course enrollment:
  Hallucinates non-existent courses despite using knowledge base results

An Example of a Conversation

Using Genie Worksheets

You 01:01 AM

Hi, I want to complete the significant implementation requirement. What courses would you suggest me to take?

Took 3 steps

Course Enrollment Assistant 01:01 AM

I found some courses that fulfill the significant implementation requirement. Here are a few suggestions along with their average hours spent per week:

1. **General Game Playing (CS 227B)**: A general game playing system accepts a formal description of a game to play it without human intervention or algorithms designed for specific games. Hands-on introduction to these systems and artificial intelligence techniques such as knowledge representation, reasoning, learning, and rational behavior. Students create GGP systems to compete with each other and in external competitions. Prerequisite: programming experience. Recommended: 103 or equivalent. *Average hours spent per week: 7.48.*
2. **Logic Programming (CS 151)**: Logic Programming is a style of programming based on symbolic logic. In writing a logic program, the programmer describes the application area of the program (as a set of logical sentences) without reference to the internal data structures or operations of the system executing the program. This course introduces basic logic programming theory, current technology, and examples of common applications. Prerequisite: CS 106B or equivalent. *Average hours spent per week: 7.5.*
3. **Introduction to Computer Graphics and Imaging (CS 148)**: This introductory course in the computer graphics sequence introduces students to the technical concepts behind creating computer-generated images. Topics include the mathematical concepts of synthetic imagery, fundamentals of light and color, rasterization, and ray tracing technology. Prerequisite: Linear Algebra and familiarity with Python and debugging. *Average hours spent per week: 8.79.*

Could you please provide me with the details of the courses you want to take?

Consideration: Time Offered

You 01:02 AM

Great when are these courses offered?

Took 3 steps

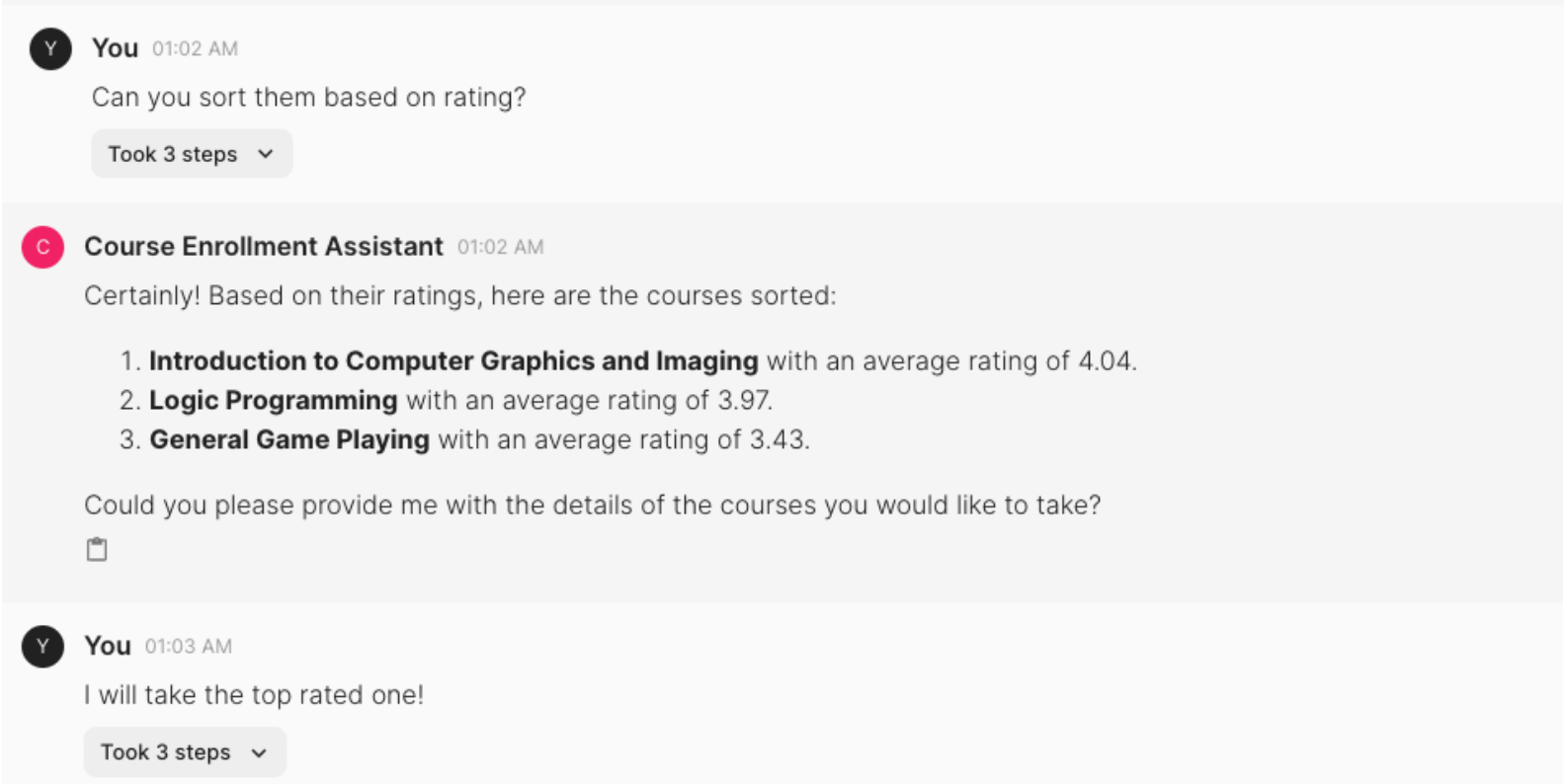
Course Enrollment Assistant 01:02 AM

The courses you're interested in are offered at the following times:

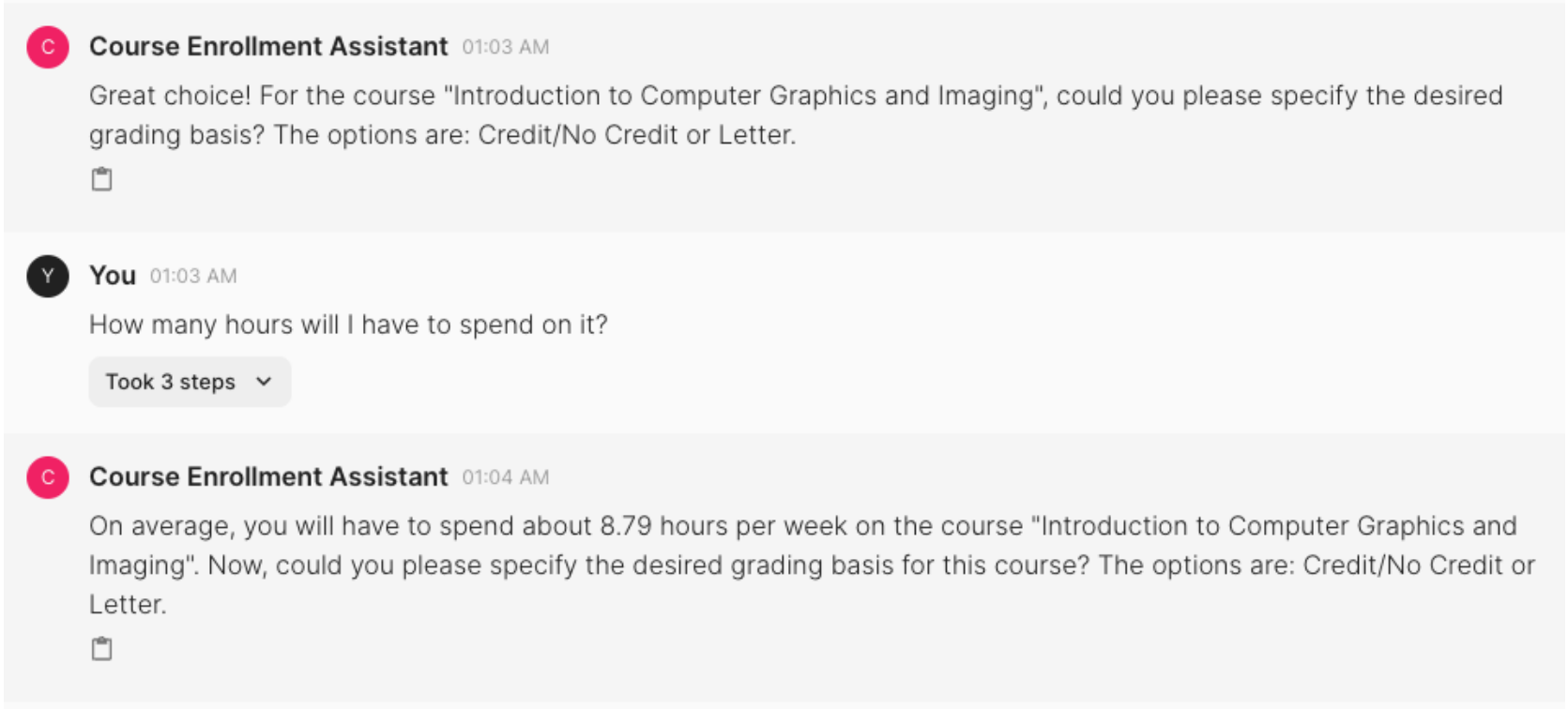
1. **Introduction to Computer Graphics and Imaging (CS 148)** is offered in the **autumn** season on **Tuesday and Thursday** from **12:00 PM to 1:20 PM**.
2. **Logic Programming (CS 227B)** is offered in the **spring** season on **Tuesday and Thursday** from **1:30 PM to 2:50 PM**.
3. **General Game Playing (CS 151)** is offered in the **spring** season on **Wednesday** from **4:30 PM to 6:20 PM**.

Could you please provide me with the details of the courses you would like to take?

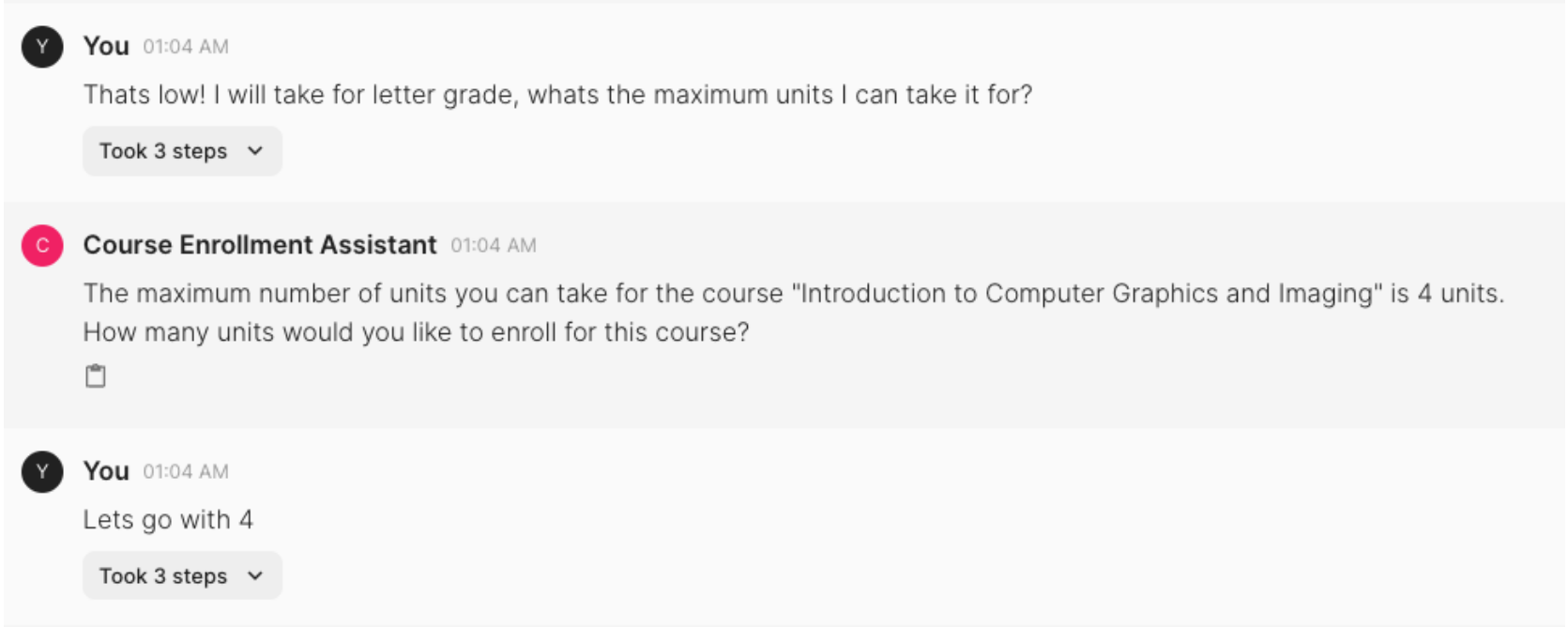
Consideration: Sort by Course Rating



Credit or Letter? Depends on the Workload



Pick Max. Number of Units



What makes it hard to implement task oriented agents?

1. Task agents are also knowledge agents

	Structured	Free Text	Tasks
Restaurants	Cuisines, opening hours, address, rating	Reviews, popular dishes	Book restaurant
Products	Product ID, cost, ratings, sizes, physical dimensions, color	Descriptions, reviews	Purchasing Ordering food
Courses	Class number, time offered, instructor, building, pre-requisites	Descriptions, reviews	Enrolling in courses
Business processes	Receipts, statements	Regulations	Filing taxes Reimbursements Customer support
Medical	Patient demographics Prescription	Diagnosis history	Making diagnoses Insurance claims

2. Mixed Initiatives

- Knowledge agents
  - Users take the initiatives in asking questions

- Task agents need to take initiatives
  - Ask users for required parameters;
    users may not answer the agents' questions
  - Suggest new actions

3. Scale: Long instructions, conversations, contexts

Long conversations!
Need to track: instructions, user requests, task parameters

What is the solution?

Common Approach: Multi-agents

A coordinator calls different agents for different functions

Advantages:

- Shorter prompt, shorter conversation context per agent

Disadvantages

- As a user, how would you like to be passed from one agent to another?
- How to answer related questions or solve simultaneous constraints?
  - E.g. hotel reservation needs to know the flight booked

Summary: Challenges of Task-Oriented Agents

1. Also a knowledge agent
2. Mixed initiatives:
   responsive to user requests, must accomplish its tasks
3. Scale: long instructions, contexts
4. SAFETY: chatbots are user-facing – can we trust it?
   We need to specify the policies:
   what the agent should do under all circumstances!

We need FULL SPECIFICATIONS and to ENFORCE adherence!

There is no free lunch.

Outline

- Challenges of task-oriented agents
- Genie Worksheets
  - **Design and Implementation**
  - Capabilities
  - Evaluation

Controllable and Reliable Knowledge-Intensive Task-Oriented Conversational Agents with Declarative Genie Worksheets

Harshit Joshi **Shicheng Liu** **James Chen** **Robert Weigle** **Monica S. Lam**
Computer Science Department
Stanford University, Stanford, CA
{harshitj, shicheng, lam}@cs.stanford.edu

https://ws.genie.stanford.edu/

https://github.com/stanford-oval/genie-worksheets

In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Vienna, Austria, July 2025.

Genie Worksheet Design Principles

Genie Worksheet: Full declarative, specification
Knowledge corpora; Tasks

- What to do under all circumstances
- What state to keep for the conversation

Genie runtime:

- Use LLMs to parse user input (with a customized context)
- Save the conversation state and perform actions based on the Worksheets
- Tell LLMs the response to generate (with a customized context)

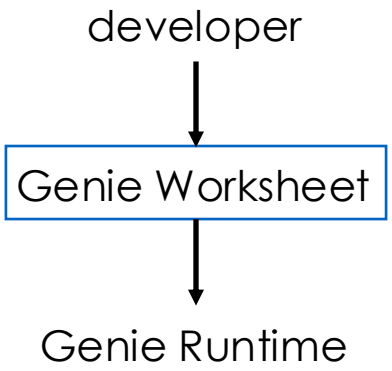
1. Knowledge agent

2. Mixed initiatives

3. Scale

4. Safety

Developer Specification



Design principle: Declarative

- Users just provide declarative info about the resources
- Genie Runtime handles the conversation flow
- Minimal information to guarantee safety

2 kinds of worksheets

- Knowledge Worksheets
  - Schemas of structured and unstructured data. (SUQL used)
- Task Worksheets
  - APIs: Function calling parameters
  - Policies: Actions in Python

1. Course Assistant Knowledge Corpus

courses DB

offerings DB

programs DB

ratings DB

	internal; primary	int	rating_id	
	internal	int	course_id	
	internal	List[str]	instructor_names	
	internal	int	average_rating	
	internal	int	num_ratings	
	internal	int	term_id	
	internal	int	start_year	
	internal	int	end_year	
	internal	Enum	season	
				autumn
				winter
				spring
				summer
	internal	List[str]	reviews	

	internal	str	sheet_requirements	

2. Stanford Course Enrollment Form

Form Name	Kind	Type	Name	Enum Values	Description	WS Action
Main	WS		course_enrollment			Say(enroll(StudentInfo, courseToTake))
	input	CourseToTake	course_to_take		The course to enroll	
	input	StudentInfo	student_info_details		Information on the student	
StudentInfo		worksheet				
	input	str	student_name		Name of the student	
	input	str	student_id		Student's ID number	
	input	str	student_email_address		Student's email address	
CourseToTake		worksheet				
	input	str	course_name		Name of the course	
	input	Enum	grade_type		The desired grading basis	
				Cred/No Cred	Taken for credits	
				Letter	Taken for grades	
	input	int	num_units		The number of units taken	
	input	confirm	confirm		Confirm the course	

Predicates needed for data dependent fields

- Some fields are needed only for some users

Agent: Are you an international student?

User: yes, I am currently on a study visa.

Agent: What type of visa is it? F1 or J1?

+ Data-Dependent Worksheets/Fields
+ Field actions

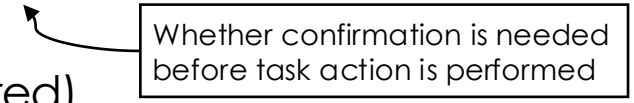
Ticket Submission Worksheet

WS Predicate	WS Name	Predicate	Kind	Type	Name	Enum Values	Description	Don't Ask	Required	Confirmation	Actions	WS Actions
	Main		**Task**									**>say(submit_ticket(self.student_task, ...))**
			input	Enum	**student_task**		The type of student requ...		TRUE			
						TroubleShoot						
						Leave of Abs.						
						Test Credits						
		self.student_task == "TroubleShoot"	input	Trouble Shoot	trouble_shoot		The enrollment issues that the student is facing		TRUE			
						• • •						
		self.student_task is not None	input	str	extra_details		Ask for any other detail that the student wants to add		FALSE			
			input	confirm	confirm		Confirm that the student wants to submit the ticket.		TRUE		if self.confirm == False: >say("Thank you, how else can I help you?") > exitws()	
self.student_task == "Trouble..."	**TroubleShoot**		**WS**									
						• • •						
	services_general_info		**KB**									

Task Worksheets

A task is like a function, with predication

- Task Definition: Name, Predicate (when is it active), Kind (Task or DB),
  Action (result of task)
- Field Definition: Name, Predicate, Kind (input, output, internal)
  Type (WS name, str, enum, …, confirm),
  Description,
  Don't Ask (Capture if info is offered),
  Required (Ask but may be optional),
  Action (code to execute on input)



- Built-in actions: say, propose
- User-define actions as python code

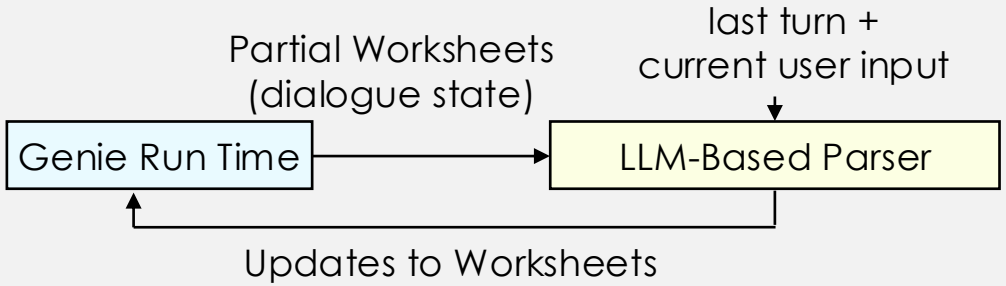
Discussion

What can Genie Worksheets do?

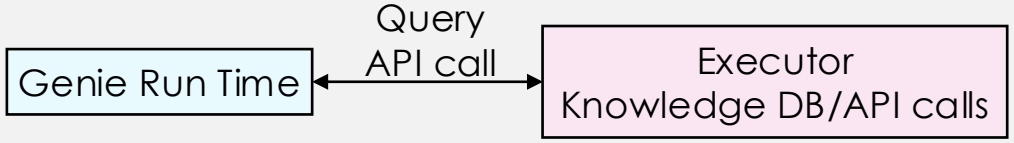
Or not do?

How Genie Handles Long Conversations

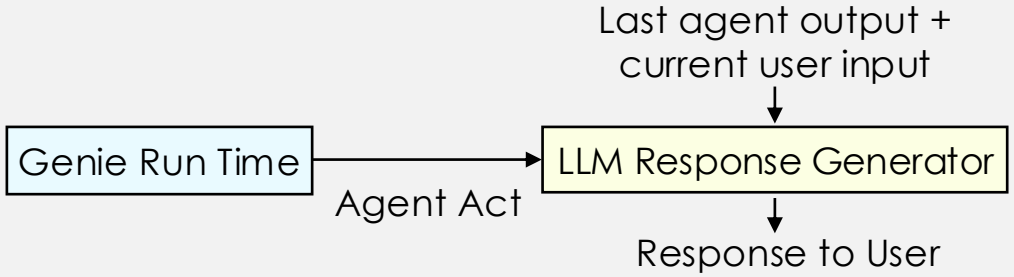
INPUT



EXECUTE POLYCY
- Queries
- Activated actions



OUTPUT
- Responses from above
- Asks for confirmation
- Asks for missing parameters in the worksheet



Example: One Conversational Turn

d_t: dialogue state at time t
u_t: user utterance at time t
$\bar{a}_t$: agent act at time t
a_t: agent utterance at time t

Student: Can you suggest courses where students are happy with the load? u_{t-1}

```
answer_1 = Answer("show courses with light workload")

answer_1.result = [{course_id: "CS161"}, {course_id: "HCI347"}, ...]

course_1 = Course(course: answer_1.result)

course_to_take_1 = CourseToTake(course_0_details=course_0)

main_1 = Main(courses_to_take=course_to_take_1)
```

d_{t-1}

Assistant: I have found the following courses that where students are happy with the workload .... Which course would you like to take? a_{t-1}

Student: I will take the HCI course. Can I take it for more than 3 units? u_t

Parser

```
answer_1 = Answer("""What is the max number of
    units for HCI 347""",
 """SELECT max_units, course_id FROM courses
 WHERE course_id ='HCI347'""")
```

Executor

d_t

```
answer_1.result = [{course_id:hci347,max_units:5}]
courses_to_take_1 = CoursesToTake(course=course_1)
course_1 = Course(course=CourseRecrod("HCI347"))
```

$\bar{a}_t$

```
[Report(answer_1, answer_1.result), AskField(course_1
 course_units)]
```

Response Generator

Assistant: Absolutely! HCI 347 can be taken up to 5 units. Would you like to take it? a_t

Outline

- Challenges of task-oriented agents
- Genie Worksheets
  - Design and Implementation
  - **Capabilities**
  - Evaluation

Managing Mixed Initiatives

USER:	ASSISTANT
I want to transfer money.	
	Sure. How much money do you want to transfer?
How much money do I have in the account?	
	You have $157.00 How much money would you like to transfer?

User Changing Previous Answers

USER: ASSISTANT

I want to fly from New York to Los Angeles on 8th this month.

I have found 3 flights for you. ... Which one would you pick?

Oh wait, I have a meeting on 8th, I think I will leave on 9th.

Support Multiple Knowledge Queries

USER: ASSISTANT

Can you suggest some highly rated NLP courses? Also, when does the next semester start?

CoursesDB (
SELECT * FROM courses WHERE
answer("is this NLP course?", description) AND ratings > 4.5)

CoursesDB(answer("when does the semester start?", ,,,)

Composition of Queries & Function Calls

Use the result of a worksheet as input to another worksheet

User: I want to book an Italian restaurant in NYC
for two on Valentine's day.

```
BookRestaurant(restaurant=
    RestaurantsDB ("SELECT * FROM restaurants WHERE
            'italian' = ANY(cuisines) AND location = 'NYC;')"),
    date="02/14/2024", and num_people=2)
```

Another Example

USER: ASSISTANT

I want to take an AI course for 3 units and letter grade.

```
Enroll(
    course=CoursesDB (“SELECT * FROM courses WHERE
       answer(description, “is the course on AI”)=“yes””),
    num_units=3, grade_type=Letter)
```

I can help you with enrolling in an AI course. Here are some top-rated AI courses: CS 224V, CS 224N, CS205. Which one would you like to take?

Summary: Genie Worksheets Can

- Support data-dependent parameter queries
- Support multiple queries and requests in a single utterance
- Support arbitrary interleaving of queries & action requests
- Keep track of essential information in long conversations
- Allow unhappy paths, eg. User changing their answers

Developer only needs to supply the Genie Worksheets

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Evaluation

- To compare with existing work: STARV2 Dataset (2022)
  - Wizard-of-Oz: 3 hardest domains--bank, trivia, trip
  - Annotated with: “user and agent dialogue acts”

Mosig, Johannes EM, Shikib Mehri, and Thomas Kober. "Star: A schema-guided dialog dataset for transfer learning."
Zhao, Jeffrey, et al. "Anytod: A programmable task-oriented dialog system."

STARv2 Evaluation

- SOTA: AnyTOD (**AT XXL**)
  - finetuned T5 (13B) on 6000 examples (all domains but the test)
  - Implement programs to handle logic
- Genie:
  - Semantic parsing prompt: 3 examples
  - Agent Policy: 9 lines in a Worksheet

System Action F1: The next agent action

Agent	Bank	Trip	Trivia
Finetuned T5 (11B)			
AT XXL	54.3	52.4	73.8
AT-SGD XXL	53.1	51.5	81.1
AT-PROG XXL	61.0	60.8	73.7
AT-PROG +SGD XXL	65.0	62.9	86.3
Zeroshot			
Llama 3.1 70B (FC)	48.9	41.7	81.7
GPT-4o-mini (FC)	50.8	43.8	69.8
GPT-4 Turbo (FC)	55.1	42.7	82.5
Genie + Llama 3.1 70B **(Ours)**	82.1	75.9	82.2
Genie + GPT-4o-mini **(Ours)**	**82.5**	80.5	90.3
Genie + GPT-4 Turbo **(Ours)**	**82.5**	**83.4**	**92.7**

Most errors caused by inconsistent data annotations

Evaluation

- To compare with existing work: STARV2 Dataset (2022)
  - Wizard-of-Oz: 3 hardest domains--bank, trivia, trip
  - Annotated with: "user and agent dialogue acts"
  - WOZ benchmarks:
    - not realistic -- few unhappy paths
    - too easy for LLMs (simple slot fills)
    - Worse annotations than LLM outputs!

LLM out-ran traditional NLP evaluation!

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Quiz
What should we do next?

Evaluation with real users

Three diverse applications with varying complexities.

- **Restaurant Reservation**: Uses the real-life database from Yelp
- **Ticket Submission**: A subset of Service Now api
- **Course Enrollment**: Uses multiple real-life databases of courses at Stanford University.

Applications	Task WSs	KB WSs	Fields	Predicates	Actions
StarV2 (Bank)	3	0	10	4	4
StarV2 (Trip)	2	0	6	0	2
StarV2 (Trivia)	2	0	6	0	3
Restaurant Reservation	2	2	19	2	3
Course Assistant	4	4	52	3	1
Ticket Submission	7	1	29	18	2

Note the difference in complexity

Baseline and Study Design

- Compare to GPT-4 turbo with Function Calling.
  - Gives GPT-4 the same KB-Parser as Genie Worksheet
- Users
  - Restaurant Reservation and Ticket Submission:
    - 22 and 20 users from Prolific crowdsource platform
  - Course enrollment:
    - 20 university students
  - Randomly assigned Genie Agent or GPT-4 (FC)

Real User Evaluation

Goal Completion Rate

	Restaurant	Course Enrollment	Ticket submission
GPT4: function calling + Genie DB parser	**54.5**	**10.0**	**0.0**
Genie Worksheet	**91.6**	**80.0**	**80.0**

GPT-4 results are not acceptable

- Low completion rate: even 54.5% on restaurant is not good enough!
- Hallucinates non-existent courses despite using knowledge base results

Genie Worksheets – deployable with additional engineering

- Only LLM-based technique that does not hallucinate on Query results/Actions

Summary:
Genie Worksheet Design Principles

Genie Worksheet: Full declarative, specification
Knowledge corpora; Task signatures; Policies

- What to do under all circumstances
- What state to keep for the conversation

Genie runtime:

- Use LLMs to parse user input (with a customized context)
- Save the conversation state and perform actions based on the Worksheet
- Tell LLMs the response to generate (with a customized context)

1. Knowledge agent

2. Mixed initiatives

3. Scale

4. Safety